

1. Decomposition

- Defining the classes (Plant and zombie classes), Initializing the instances (2 plant objects and 1 zombie object), Writing the turn loop, creating the health and distance displays, and checking who won/loss.

2. Pattern recognition

- Each living plant attacks in order.
- Health value and the distance of the zombie is displayed after each attack.
- The game checks whether either side has won.

3. Abstraction

Plant

- Name
- Health
- Damage

Zombie

- Name
- Health
- Damage
- Distance

4. Algorithm design

- Create a Plant class
- Create a zombie class
- Give each object its own attributes (name, health, damage, and/or distance)
- Create two plant objects with different damage values.
- Create one zombie object.
- Make the plants attack the zombies.
- Make the zombies move toward the plants and attack when the distance reaches zero.
- Make the zombie target the next plant when the first one is dead
- Check the health of all objects after every action.
- Stop the game when either the zombie or both plants are dead.
- Display the final stats and the winner.